

L1.4 The Derivative

Work in your group. Check your answers against the Quarto tonight.

1 · The derivative at a point

Stop. Write the difference quotient from memory for recall purposes.

1. $f(x) = 2x^2 + 1$. Find $f'(2)$

2. What is the **equation of the tangent line** to $f(x) = 2x^2 + 1$ at $x = 2$.

What extra piece of information did you need?

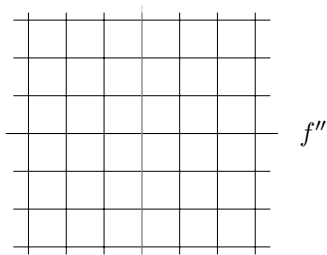
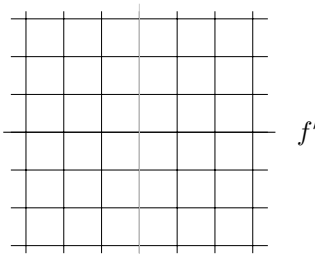
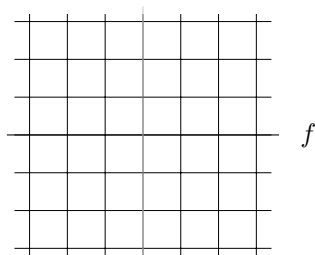
3. A particle's position is $x(t) = 5t^2$.

- (a) Find its **average** velocity on the interval from $t = 5$ to $t = 5 + h$. Leave h in your answer.
- (b) Now find its **instantaneous** velocity at $t = 5$.
- (c) Which one of those is a secant slope and which is a tangent slope?

2 · The derivative as a function

4. Find $\frac{dy}{dx}$ for $y = \sqrt{x}$ **from the definition of a derivative**, then evaluate $\left. \frac{dy}{dx} \right|_{x=3}$.

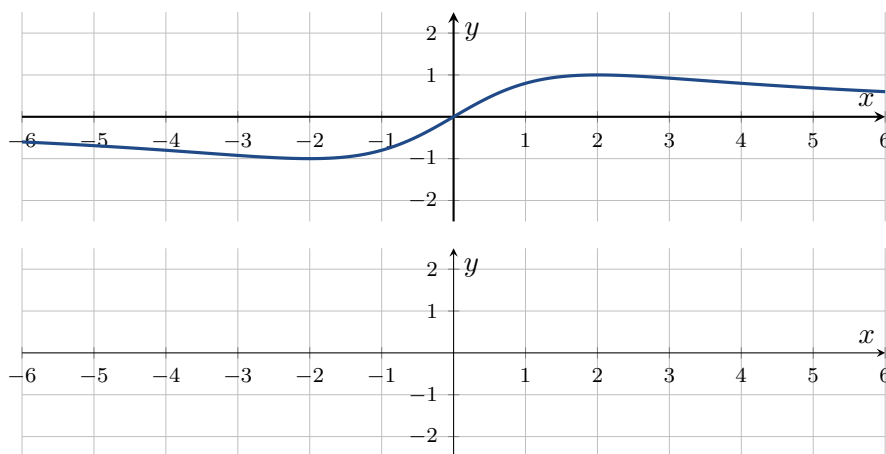
5. $f(x) = x^2 - 4$. Find $f'(x)$ and then $f''(x)$. Sketch all three stacked on the grids below.



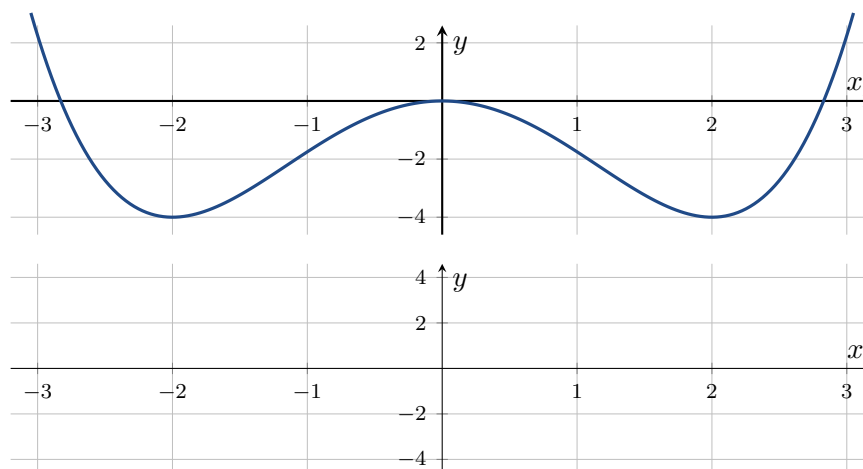
Stop. What does a horizontal tangent on f become on f' ? Write it down before you sketch.

6. Sketch f' on the empty grid.

Where does f have HTs, and IPs? What are the intervals its increasing and decreasing? And how is that related to the sign of f' ?



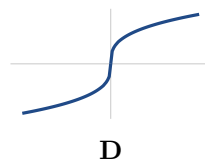
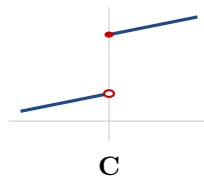
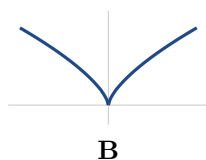
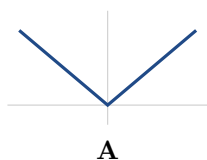
7. Sketch f' . Be sure to mark IPs and HT points.



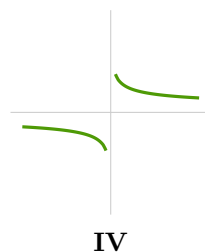
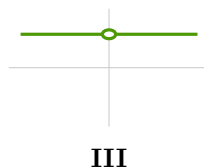
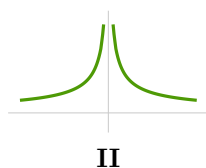
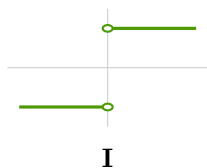
3 · Differentiability

8. Match each graph of f to the graph of its f' .

The functions.



The derivatives.



A → _____ B → _____ C → _____ D → _____

9. $f(x) = |x|$.

- (a) Write f as a piecewise function, then write f' as a piecewise function.
- (b) Show f is **not differentiable** at 0.
- (c) Is f **continuous** at 0?

10. $f(x) = |x^2 - 4|$.

- (a) Write f piecewise. *Solve* the inequalities.
- (b) Write f' piecewise.
- (c) For which values of x is f differentiable?

11. **BOARDS** Find $f'(x)$ for $f(x) = \frac{1}{x}$. Also, find the equation of the TL at $x=2$.